Project Plan
SUSTAIN 3S03 Implementing Sustainable Change

Project. Exploring Native Groundcover Alternatives to Traditional Turf Grass at McMaster University

Community Project Champion (CPC). Dr. Susan Dudley, Professor, Department of Biology, McMaster University

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Section 3.2 shows how each objective and its deliverables connect back to the problem described in section 1.

1. Problem Statement

Turf grass is the default surface of most campuses, parks, and yards, and it is costly to keep. A national estimate for the United States found that turf grass may cover an area about three times larger than any irrigated crop (Milesi et al., 2005). Keeping that grass short and green takes fuel, fertilizer, and labour, and a study of home lawns across the country estimated that carbon released by mowing and fertilizer use cancels out the carbon stored in the soil after 66 to 199 years, depending on management (Selhorst & Lal, 2013). Frequent mowing also limits what lawns offer wildlife. In Germany, areas mown twice a year held significantly more wild bee species than lawns mown twelve times a year (Wastian et al., 2016). In Massachusetts, yards mowed every three weeks had up to 2.5 times more lawn flowers than yards mowed more often (Lerman et al., 2018).

This matters because pollinators are under pressure. Declines have been reported in several bumble bee species that were historically present in Ontario, and habitat loss is one of the threats (Gibson et al., 2019). Flowering groundcover that needs little mowing could give native bees food and help them reproduce on land that is now mostly grass (Turo & Gardiner, 2021).

At McMaster University in Hamilton, Ontario, conventional turf needs frequent mowing and upkeep, especially in wet growing seasons (Dudley & Terryberry, 2026). As part of its sustainability work, the university wants to know whether native groundcover can handle clay soil, full sun, and moderate foot traffic from students, staff, faculty, and visitors (Dudley & Terryberry, 2026). The test site is a demonstration plot near the McMaster helipad, where full sun and clay soil make establishment demanding. What decision makers still need is local evidence on how native seed mixes establish on McMaster clay compared with the turf they would replace.

Our group of five students will work with Dr. Susan Dudley, a professor in the Department of Biology and our Community Project Champion. Nature @ McMaster and Facility Services will prepare the plots, and we will update Dr. Dudley every week. Friends and classmates we invite

will help on planting day. Students, staff, and visitors who walk past the site will see the plot as it grows, and native bees and other pollinators could gain a new source of food. Faculty of Science students who may continue monitoring the plot from spring 2027 will also rely on the records we leave behind.

We can help by reviewing the research, choosing species with a clear scoring framework, assessing three campus green spaces, seeding a replicated demonstration plot in November, and recording a baseline that spring monitoring can be compared against. That work will help campus staff decide whether native groundcover is a realistic alternative to turf in places where mowing is costly. This is why we are pursuing this project throughout the term.

2. Objectives

We have three objectives, and each one builds on the last. The first finds the species, the second tests where on campus they could work, and the third turns what we learn into guidance McMaster can use.

No.	Objective
1	Identify six suitable native groundcover species through research by October 30, 2026.
2	Assess three McMaster campus green spaces for potential native groundcover suitability by November 28, 2026.
3	Recommend three groundcover species and future monitoring strategies in a report by December 4, 2026.

2.1 How each objective meets the SMART criteria

Each objective was checked against the SMART criteria in the course rubric, which ask for goals that are specific, measurable, achievable, relevant, and time bound. The idea comes from an earlier management checklist that asked for goals that are specific, measurable, assignable, realistic, and time related (Doran, 1981).

	Objective 1	Objective 2	Objective 3
Specific	Six species native to Ontario that tolerate clay soil, full sun, and moderate foot traffic, need little or no mowing, and support pollinators.	Three campus green spaces, including the helipad demonstration plot, assessed for sun, soil, drainage, slope, foot traffic, and current mowing.	Three of the six species ranked for McMaster, plus a plan for monitoring germination, percent cover, weeds, mowing needs, and pollinator visits.

	Objective 1	Objective 2	Objective 3
Measurable	Six species scored with the evaluation framework, each backed by at least two sources from a log of 15 or more.	One completed assessment sheet with dated photos for each space, each space rated against the six species, and the helipad plot seeded.	One final report with three ranked species and reasons, a monitoring protocol, and a blank data sheet.
Achievable	Five members each log at least three sources over three weeks, at about four hours a week.	Dr. Dudley helps choose the spaces, Nature @ McMaster prepares the helipad plot, and visits fit into weeks 5 to 9.	Writing starts in week 10, and Dr. Dudley sees a draft on December 1.
Relevant	The criteria come straight from the project description, and the list feeds both the site assessments and the final recommendations.	It shows where native groundcover could realistically replace turf on campus, not just in theory.	It gives McMaster a clear next step and lets future students continue monitoring the helipad plot.
Time bound	Shortlist approved by Dr. Dudley by Friday, October 30, 2026.	All three assessments done and the helipad plot seeded by Saturday, November 28, 2026.	Report delivered to Dr. Dudley by Friday, December 4, 2026.

3. Deliverables

Each deliverable below is a document, a planted plot, or a presentation that will exist by December 5, 2026. Every objective has at least two deliverables tied to it.

No.	Deliverable and what it contains	Objective	Due	Lead
D1	**Literature review.** At least 15 primary sources in APA format on four themes, which are the costs of turf maintenance, the value of lawn alternatives for pollinators, how native groundcover handles clay soil and foot traffic, and seeding practice for late autumn.	1	October 30	Joseph

No.	Deliverable and what it contains	Objective	Due	Lead
D2	**Evaluation framework and species table.** A weighted scoring table for clay tolerance, full sun tolerance, foot traffic tolerance, mowing need, Ontario native status, pollinator value, bloom timing, and seed cost and availability. It includes the referenced comparison of candidate species with their advantages and disadvantages, a shortlist of six species with a written reason for each, and a ranked backup list of substitute species.	1	October 30	Marchelino
D3	**Site assessments and planting plan.** An assessment sheet with dated photos for each of the three green spaces, covering sun, soil, drainage, slope, foot traffic, and current mowing, with each space rated against the six species. For the helipad plot it adds a labelled plot map with twelve subplots in three blocks, the random order of treatments in each block, the species and seeding rate for each mix, the amount of straw, the site preparation agreed with Nature @ McMaster, a supply list, a schedule for planting day, and the weather, cold, and helipad safety rules.	2	November 13	Mehrshid
D4	**Seed purchasing summary and seed record.** A table of suppliers, stock, package sizes, prices, and delivery dates, the seed bought for all three mixes, and a record of each species, amount, seed lot, and origin so that future researchers can trace what was sown.	2	November 13	Marchelino
D5	**Planting event and seeded plot.** The nine native subplots seeded, raked, and covered with straw, and the three turf subplots marked. The event has a sign in sheet that records photo consent, and photos. The planned date is Saturday, November 21 and the backup is Saturday, November 28.	2	November 28	Mehrshid

No.	Deliverable and what it contains	Objective	Due	Lead
D6	**Baseline and establishment record.** Photos of the site before preparation and of every subplot after seeding, GPS points for every subplot corner, a final plot map with measurements, site notes on sun, slope, and soil, and the sowing record with the date, species, and amounts.	3	December 1	Diya
D7	**Species recommendations and monitoring strategy.** Three recommended species ranked with reasons, and a protocol for spring and summer 2027 and later years that covers germination counts, percent cover, weed cover, mowing needs, and pollinator visits, with a blank data sheet.	3	December 4	Diya
D8	**Final report and handover.** One report that joins D1 to D7, explains its limits, and is delivered to Dr. Dudley on December 4 with all files placed in a folder that Dr. Dudley owns, then submitted on Avenue on December 5.	3	December 4	Hadar
D9	**Final presentation.** A slide deck that summarizes our findings and recommendations, finished by December 4, presented on [presentation date], and submitted on Avenue on December 5. Each member presents the part they led.	3	December 4	Joseph

3.1 Implementation overview

We will compare three native seed mixes with a turf grass control on the demonstration plot near the McMaster helipad. Each of the four treatments will be repeated in three blocks, which gives twelve subplots. Repeating treatments matters because clay, drainage, and compaction can change across even a small plot, and replication lets us tell a real difference between mixes from a patch of unusual ground. Within each block the order of treatments will be drawn at random and recorded in the planting plan. The turf control is the existing campus turf left in place in each block, so Nature @ McMaster only needs to prepare the nine native subplots. In Cleveland, vacant lots seeded with a native wildflower mix or with fine fescue grass were linked with more larvae of native cavity nesting bees and wasps (Turo & Gardiner, 2021). That result is one reason we are testing seeded mixes and not just mowing turf less often.

Dr. Dudley is still discussing the site with grounds staff, so we will confirm the plot size, subplot size, and exact location with Nature @ McMaster and the grounds team at a site visit in week 5 before we lock the design, and we will choose the other two green spaces to assess with Dr. Dudley at the same time. At that visit we will also ask how the turf will be removed, on what date, and how deep, and we will photograph the site before that work starts so the baseline includes the original turf. Because the plot is near the McMaster helipad, we will ask how far it must sit from the landing area, whether loose straw is allowed there, and who we should notify before planting day.

Nature @ McMaster and Facility Services will prepare the plots and Dr. Dudley will supply the straw. We will weigh the seed for each subplot into a labelled bag, mix it with dry sand so it spreads evenly, broadcast it by hand, rake it lightly into the surface, and cover it with a thin layer of clean straw that is free of weed and grain seed. Seeding in the second half of November puts the seed on the ground as the growing season ends, so it can sit through winter and germinate in spring. In tallgrass prairie restoration trials, the authors recommended broadcasting seed in the dormant season over broadcasting in the growing season (Larson et al., 2011). Our late November date puts us in that dormant window.

Nothing will have germinated by December 5, so what we can show this term is the research, the plot design, and a documented baseline. Germination counts and percent cover will be recorded in spring 2027 by whoever continues the monitoring, which may include a Faculty of Science thesis student. For that reason we will write the monitoring protocol and data sheet in November so they are ready to use.

3.2 How the plan fits together

Each of our three objectives responds to a gap we describe in the problem statement. The first gap is that campus staff do not yet know which native species can handle McMaster clay, full sun, and foot traffic. Objective 1 deals with this through the literature review that Joseph leads and the evaluation framework that Marchelino leads (D1 and D2) in weeks 4 to 6.

Knowing which species should work is not the same as seeing them grow on campus, so Objective 2 assesses three campus green spaces and puts the shortlist into the ground at the helipad plot beside a turf control. Mehrshid leads the site assessments, the plot design, and the planting (D3 and D5) in weeks 5 to 10, Marchelino handles the seed (D4), and all five of us plant.

Even a well planted plot is only useful if someone knows what to measure next spring. Objective 3 covers that with the baseline record and the three species recommendations and monitoring strategy (D6 and D7), which Diya leads, the final report (D8), which Hadar compiles, and the presentation (D9), which Joseph coordinates, in weeks 10 and 11. Section 4 places each of these deliverables in a specific week.

4. Timeline

The timeline covers every week from September 21 to December 5, 2026. The chart gives the overview, and the weekly table lists each task, meeting, email, deadline, and the average hours each member should plan for.

4.1 Timeline at a glance

Work stream	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11
Weekly group meeting											
Weekly meeting with Dr. Dudley											
Weekly email to Dr. Dudley (graded)											
Project plan writing and submission											
Literature review (D1)											
Evaluation framework and species table (D2)											
Green space assessments (D3)											
Seed quotes, order, and delivery (D4)											
Plot design and planting plan (D3)											
Planting event and backup date (D5)											
Baseline and establishment record (D6)											
Species recommendations and monitoring (D7)											
Final report (D8)											
Final presentation (D9)											

Dark green shows planned work. Light green shows work already completed. Week dates are listed in the week by week plan that follows.

4.2 Week by week plan

Week	Dates	Tasks	Meetings and emails	Deadlines and milestones	Hours per member
1	Sep 21 to 27	• Group formed, with a Teams chat and a shared folder set up. • Project expectations reviewed and first reading on clay tolerance, full sun, foot traffic, low maintenance, less mowing, and pollinator value. • Research areas divided among members.	• Group meeting 1 to set roles and a research plan. • Introductory weekly email to Dr. Dudley.	Completed. Research areas assigned.	2
2	Sep 28 to Oct 4	• Preliminary species list and evaluation criteria started. • Plan sections divided and drafting begins. • Group contract drafted.	• Group meeting 2 on Tuesday, September 29 at 1.30 p.m. • Weekly email asking which species to prioritize.	Completed. Preliminary criteria listed.	3
3	Oct 5 to 11	• Group contract signed on Monday, October 5, and signed again after final edits. • Plan finished, checked against the rubric, and edited into one voice. • Plan emailed to Dr. Dudley with all members copied, and the screenshot added.	• Group meeting 3, our first meeting with Dr. Dudley, on Wednesday, October 7 at 10.45 a.m., which set our weekly time. • Dr. Dudley meets grounds staff about the site on Thursday, October 8. • Weekly email.	**Project Plan due Friday, October 9, by 11.59 p.m.** Hadar uploads the PDF to Avenue.	6

Week	Dates	Tasks	Meetings and emails	Deadlines and milestones	Hours per member
4	Oct 12 to 18	• Monday, October 12 is Thanksgiving, so this is a lighter week. • Literature review begins and the group logs its first eight sources in the shared source log. • Draft evaluation criteria listed. • Each member posts their November test and assignment dates in the timeline tracker by Friday, October 16.	• Group meeting 4 with Dr. Dudley on Wednesday, October 14, where we ask about the seed budget, who pays, and how the order is placed. • Weekly email.	Source log reaches 8 sources by Friday, October 16.	3
5	Oct 19 to 25	• Mehrshid walks the helipad site on Monday, October 19, records sun, slope, drainage, and soil, and photographs the existing turf before preparation. • Site visit with Nature @ McMaster by Friday, October 23 to confirm the helipad plot size, preparation, and helipad rules, and the other two green spaces chosen with Dr. Dudley. • Reading finished at 15 or more sources and review sections drafted. • Evaluation framework finalized after CPC feedback.	• Group meeting 5 with Dr. Dudley on Wednesday, October 21, where the criteria and weights are reviewed. • 15 minute online check before the October 30 deadline. • Weekly email.	Literature review draft and framework ready by Friday, October 23.	5

Week	Dates	Tasks	Meetings and emails	Deadlines and milestones	Hours per member
6	Oct 26 to Nov 1	• Candidate species scored and six chosen. • Assessment sheets completed for the other two green spaces. • Seed quotes, stock, and delivery dates collected from at least two suppliers. • Volunteer waiver and McMaster approval question sent to Dr. Dudley by Friday, October 30. • Literature review finalized.	• Group meeting 6 with Dr. Dudley on Wednesday, October 28, where the shortlist is approved. • Weekly email.	**Objective 1 due Friday, October 30.** D1 and D2 complete.	5
7	Nov 2 to 8	• All three green spaces rated against the six species. • Plot layout and the three mixes drafted from the confirmed plot size. • Final seed cost checked against the budget Dr. Dudley gave in week 4.	• Group meeting 7 with Dr. Dudley on Wednesday, November 4, where the three mixes and seeding rates are approved. • 15 minute online check before the November 13 deadline. • Weekly email.	Seed order placed by Wednesday, November 4. Green space assessments complete by Friday, November 6.	4

Week	Dates	Tasks	Meetings and emails	Deadlines and milestones	Hours per member
8	Nov 9 to 15	• Design and planting plan finished. • Nature @ McMaster prepares the nine native subplots on the date agreed at the site visit. • Each lead outlines their report section by Friday, November 13. • Hadar asks for written confirmation of plot readiness and straw delivery on Monday, November 9. • Seed delivery checked and recorded. • Three to five volunteers invited and planting day tasks assigned.	• Group meeting 8 with Dr. Dudley on Wednesday, November 11, where the planting plan and the November 28 backup date are approved and we ask who will monitor in 2027. • 15 minute online check before planting. • Weekly email.	D3 and D4 complete by Friday, November 13.	4
9	Nov 16 to 22	• Plot readiness and straw delivery confirmed in writing by Monday, November 16. • Forecast checked daily, with a written weather decision on Wednesday, November 18. • Seed weighed into labelled bags and subplots marked. • **Planting event on Saturday, November 21 from 10 a.m. to 1 p.m.**	• Group meeting 9 with Dr. Dudley on Wednesday, November 18. • Weekly email.	Planned planting date, Saturday, November 21.	6

Week	Dates	Tasks	Meetings and emails	Deadlines and milestones	Hours per member
10	Nov 23 to 29	• Backup planting on Saturday, November 28 only if November 21 is cancelled. • Debrief within 48 hours of planting. • Baseline photos, GPS points, and sowing record compiled. • Monitoring protocol, report sections, and slides drafted.	• Group meeting 10 with Dr. Dudley on Wednesday, November 25. • 15 minute online check before the December 4 deadline. • Weekly email.	**Objective 2 due Saturday, November 28.** D5 complete. Four reserve hours each if the backup date is used.	4
11	Nov 30 to Dec 5	• Monday, November 30 report draft finished. • Tuesday, December 1 baseline record final and draft sent to Dr. Dudley, with comments requested by noon on Thursday, December 3. • Thursday, December 3 CPC comments added and presentation rehearsed.	• Group meeting 11 with Dr. Dudley on Wednesday, December 2. • Final weekly email.	**Friday, December 4.** Report, recommendations, and handover folder delivered to Dr. Dudley, and slides finished. All project activity ends. **Saturday, December 5.** All deliverables submitted on Avenue.	5

Weeks run from Monday to Sunday, except week 11, which ends on Saturday, December 5. The hours column is the average planned workload for each member, including about one hour of meetings, and section 5.3 shows how it splits by person. The load is lightest in weeks 1, 2, and 4, the last of which includes the Thanksgiving holiday. It is heaviest in the plan week and the planting week, at six hours each. Over eleven weeks that adds up to about 47 hours for each member.

4.3 Key dates and assignment due dates

Date	Milestone or group assignment due date
Every week	Weekly email to Dr. Dudley with all members copied, covering the work completed, challenges, questions, next steps, and a draft agenda. It is due [add the official weekly due day from the course outline].
Tuesday, September 29	Group meeting 2 at 1.30 p.m., where plan sections were divided.
Monday, October 5	Group contract signed, and signed again after final edits before submission.
Wednesday, October 7	First meeting with Dr. Dudley, which set our weekly Wednesday time of 10.45 a.m.
Friday, October 9	Project Plan due on Avenue and emailed to Dr. Dudley by 11.59 p.m.
Friday, October 30	Objective 1 complete with the literature review (D1) and the evaluation framework and species table (D2).
Wednesday, November 4	Seed order placed.
Friday, November 6	Assessments of all three green spaces complete.
Friday, November 13	Site assessments and planting plan (D3) and seed purchasing summary and seed record (D4) complete.
Saturday, November 21	Planned planting event (D5). Backup date is Saturday, November 28.
Saturday, November 28	Objective 2 complete with three green spaces assessed and the helipad plot seeded and mulched.
Tuesday, December 1	Baseline and establishment record (D6) final and report draft sent to Dr. Dudley.
Friday, December 4	Final report, recommendations, and handover delivered to Dr. Dudley (D7 and D8), and slides finished (D9). All project activity ends.
Saturday, December 5	All deliverables submitted on Avenue for grading (D1 to D9).
[Other dates]	[Add any other group due dates from the course outline, such as the presentation day]

5. Assigned Primary Roles and Responsibilities

Each of us leads one part of the project and helps with the others, and all five of us should be able to explain any part of the plan. Each person's role follows the section of this plan they drafted, so the person who planned a part is the one who carries it out.

Member and primary role	Specific responsibilities	Why this role and how it fits with the others
Joseph DeGiorgi **Timeline and Research Coordinator**	• Builds and updates the timeline so every task, meeting, email, and deadline is in it. • Checks progress each week and adjusts the timeline when something moves. • Organizes the research and runs the shared source log. • Leads the literature review (D1). • Coordinates the final presentation (D9).	Joseph drafted the timeline for this plan, so Joseph already tracks every date and dependency. The research Joseph organizes feeds Marchelino's scoring, and the timeline depends on weekly updates from everyone.
Hadar Al-Ali **Deliverables and Communication Coordinator**	• Tracks every deliverable and checks that each one connects to an objective. • Organizes the Wednesday meetings, posts the agenda, and posts the notes within 24 hours. • Sends the weekly email to Dr. Dudley with all members copied. • Keeps the decision log and the contribution log. • Submits assignments on Avenue and compiles the final report (D8).	Hadar drafted the deliverables and the group contract, so Hadar knows what we promised and how we agreed to work. Hadar relies on progress updates from every member and works with Joseph to keep the timeline and deliverables matched.
Marchelino Toma **Objectives and Species Selection Lead**	• Keeps project work aligned with the three objectives. • Leads the evaluation framework and species table (D2). • Scores candidate species and coordinates the final shortlist with Dr. Dudley. • Prepares the seed purchasing summary, places the approved order, and keeps the seed record (D4).	Marchelino drafted the objectives, so the species work stays tied to what we set out to learn. The shortlist has to be ready before Mehrshid can finish the plot design.

Member and primary role	Specific responsibilities	Why this role and how it fits with the others
Mehrshid Abdoli **Risk and Planting Lead**	• Tracks all risks in section 7 each week, checks in with each risk owner, and leads the response to the weather and helipad risks. • Leads the three green space assessments and the planting plan (D3). • Calculates seed and straw amounts with Marchelino. • Coordinates planting needs with Nature @ McMaster and leads the planting event (D5). • Writes the weather, cold, and helipad safety rules for planting day.	Mehrshid drafted the risks and owns the two that matter most on planting day, so one person tracks the full list and leads the planting. The design depends on Marchelino's final shortlist.
Diya Jain **Community Context and Recommendations Lead**	• Keeps the work connected to the problem statement, the campus community, and the stakeholders. • Compiles the baseline and establishment record (D6). • Leads the species recommendations and monitoring strategy (D7). • Edits the final report with Hadar so it reads in one voice.	Diya drafted the problem statement, so Diya is best placed to check that the final report answers the need we described. The recommendations depend on the research, the scoring, the design, and the planting record.

5.1 Shared responsibilities

Although each member leads one area, all five of us are responsible for the whole project. We all attend the weekly Wednesday meeting and the planting event. Each member finds and logs at least three of the 15 or more sources and writes a short summary of each, so everyone understands the research behind the shortlist. Every deliverable has a lead and a second reader, and all members read each full draft at least 48 hours before it is due. The second readers are Hadar for D1, Joseph for D2, Diya for D3, Mehrshid for D4, Marchelino for D5, Joseph for D6, Hadar for D7, Mehrshid for D8, and Marchelino for D9. We update our tasks in the timeline tracker every week and raise any delay in Teams the same day. For the final presentation each member presents the part they led, but we rehearse together so any of us can answer questions on any section. All of us also keep an eye on the risks in section 7 and raise warning signs as soon as we see them.

5.2 Dependencies between members

Several tasks depend on another member's work, so we planned each handoff with a date. Marchelino needs Mehrshid's notes from the October 19 site walk and Dr. Dudley's feedback on the criteria by October 23 to finish the framework. Mehrshid then needs the approved shortlist by October 30 to build the three mixes. Marchelino needs the mixes and seeding rates approved at the November 4 meeting, and the budget Dr. Dudley gave in week 4, before placing the seed order that same day.

Planting depends on two outside partners. Nature @ McMaster prepares the plot and Dr. Dudley supplies the straw, and both need to be ready by November 18. Hadar asks for written confirmation of both on November 9 and checks again on November 16. After the event, Diya depends on everyone for photos, attendance, and measurements, which the whole group uploads to the shared folder within 24 hours. Finally, Hadar depends on Joseph, Marchelino, Mehrshid, and Diya for their report sections, which are due 24 hours before the Monday, November 30 draft.

5.3 Workload balance

We balanced the workload by hours as well as by tasks. Over eleven weeks Joseph plans about 47 hours, Hadar about 48, Marchelino about 47, Mehrshid about 47, and Diya about 46, which is 235 hours for the group. In weeks 1 to 6, Joseph and Marchelino carry more because the research and the scoring are front loaded, with 26 and 27 hours, while Hadar and Diya plan 23 each and Mehrshid plans 21. In weeks 7 to 10 the balance shifts toward the field work, so Mehrshid plans 22 hours, Hadar 19, Diya 18, Marchelino 16, and Joseph 15. In week 11 Joseph and Hadar plan 6 hours each for the presentation and the report, Diya plans 5, and Marchelino and Mehrshid plan 4. This works out to roughly four hours a week for each member, which is realistic alongside other courses.

6. Group Contract

This contract sets out how the five of us will work together from September 21 to December 5, 2026. We made it detailed on purpose, so that it still works in a busy week when someone lets things slip. A study of 239 undergraduate project teams found that a strong team charter made the biggest difference in teams whose members were less conscientious (Courtright et al., 2017).

A. Group communication

Our Microsoft Teams group chat is our main channel for updates, questions, and reminders. Our shared Google Drive folder holds one version of every file, and McMaster email is used for anything sent to Dr. Dudley or Nature @ McMaster, with all five members copied. Each member replies in Teams within 24 hours, and anyone who needs more time to reply or to finish a task says so in the chat. A message marked URGENT needs a reply within 6 hours.

Hadar sends the weekly email to Dr. Dudley with all members copied. It covers the work completed, any challenges, our questions, the next steps, and a draft agenda for the next meeting.

Nathan Butterworth, who helps with scheduling, is copied when we arrange meeting times. Meeting notes are posted in Teams within 24 hours, decisions go in the decision log, and each member posts a short status update every Friday. We stay respectful, stay open to feedback, and tell the group early about any problem that could affect our work.

B. Frequency and structure of meetings

We meet every Wednesday at 10.45 a.m., which is also the weekly time Dr. Dudley confirmed with us, so most meetings start with Dr. Dudley and then continue as group work time. Hadar posts the agenda 24 hours ahead. Each meeting reviews completed work, checks the timeline and the risks, discusses challenges, and assigns next steps. Note taking rotates each week, and Hadar posts the notes.

We also add a 15 minute online check one week before each major deadline, and an extra meeting before a CPC meeting when there is a decision to prepare. Everyone comes prepared to discuss their own tasks and any questions that need group input.

C. Attendance and participation

Every member attends every meeting and takes part in the discussion. A member who cannot come tells the group at least 24 hours ahead, reads the notes, and sends their update in writing before the meeting. Arriving more than 10 minutes late counts as a late arrival. Everyone is responsible for staying informed about the whole project, even when working on one section.

The planting event is mandatory for all five members. The only exception is an emergency or an absence the group agrees to in advance, and the absent member then takes on the debrief notes and the photo log afterward.

D. Division of work

Each member has a primary role, which is set out in section 5, and each deliverable has a lead and a second reader. Roles and hours are balanced so that every member does about the same amount of work over the term. Internal deadlines are set 48 hours before every real deadline so there is time to read, edit, and fix problems. A member who feels overloaded says so at least three days before a task is due so that we can adjust fairly, and nobody quietly drops a task. All five of us share responsibility for making sure the final project is accurate, complete, and on time.

E. Decision making

We make decisions in steps. First, each member states their view and reasons without interruption, and we try to reach agreement through discussion and compromise. If we still disagree by the end of the meeting, we vote and the majority decides, which means three of five members. A vote needs at least three members taking part, and a member who cannot attend can send their view and vote in writing before the meeting. When there are three options and none gets three votes, we drop the option with the fewest votes and vote again. If no option reaches three votes after that second round, the member who leads that area of the project makes the final

choice and records the reasons in the decision log. Anyone in the minority can add a note to the decision log.

Some decisions need more than a vote. Any change to the objectives, the deliverables, the site, the species and mixes, the budget, or the planting dates needs the agreement of all five members and a written yes from Dr. Dudley. If one of these decisions is still split after a second discussion, the member who disagrees explains what would change their mind, and we take the options to Dr. Dudley at the next CPC meeting and follow that advice. Until then the current plan stays in place. We will ask Dr. Dudley to approve the November 28 backup date during week 8, so moving to it on November 18 only needs the five of us to agree.

If something small cannot wait for the next meeting, three members may decide after the others have been told in Teams and given 12 hours to reply, and anyone can ask for it to be reviewed at the next meeting. This rule never applies to the decisions in the previous paragraph. Each lead can also make small choices inside their own role, such as the layout of the source log or the order of planting day tasks, and reports them at the next meeting. Every decision, its reasons, and the person responsible are written in the decision log within 24 hours. If a decision turns out to be a mistake, we review it openly and change it without blame. Questions about marking, course rules, or extensions go to the TA or the instructor and are not decided by a vote.

F. Procedure for reviewing and submitting assignments

The lead for an assignment finishes a draft at least 72 hours before the deadline. Each member then reviews the full assignment and not only their own section, checking it against the rubric, the word limits, the citation style, the formatting, and the accuracy of the information. Edits are finished 48 hours before the deadline. When a draft goes to Dr. Dudley for comments, the same 48 hour rule applies to that draft, and their comments are added the day before the deadline.

Before upload, each member replies in Teams that they approve the final PDF. Hadar then submits it on Avenue, with Joseph as backup if Hadar is unavailable, and posts a screenshot of the Avenue confirmation in Teams. For the project plan, the same PDF is emailed to Dr. Dudley with all members copied before the deadline, and the screenshot of that email is added to the plan.

G. Accountability when expectations are not met

A task counts as missed when it is not done by its internal deadline and the member did not warn the group at least 24 hours before. Two late arrivals or one unexplained absence count the same as one missed task. After a first missed task, another member sends a private and friendly reminder in Teams and asks whether they need help, and the member has 24 hours to finish or to agree on a new date with the group.

After a second missed task, we discuss it at the next meeting, set a revised deadline in writing, and pair the member with a check in partner. If the deadline is at risk, the group moves the task to someone else, and the member takes an equal task from the next week's list, which Hadar records in the contribution log. If problems continue after that and cannot be solved inside the

group, we contact the TA and then the instructor for guidance. The contribution log is a simple weekly record of tasks, hours, and attendance that every member can see, and we will describe each member's contribution honestly in any peer evaluation. If a member is struggling because of health, family, or other courses, we ask them to tell us early, and we will adjust the workload where we can so that nobody is penalized for being honest.

H. Raising and resolving concerns or conflicts

A member with a concern raises it early, within 48 hours of noticing it, and speaks to the person involved in private, using specific examples and I statements. A member who does not feel comfortable speaking to the person directly can skip this step and raise it at the next meeting or contact the TA. If the issue is not solved within a week, or sooner if a deadline is at risk, we set aside 20 minutes at a meeting. A member who is not involved acts as a neutral facilitator, everyone gets a chance to explain their view, and we write down one agreed solution. If most of the group is involved, we ask the TA to facilitate instead.

If the problem continues after two weeks or starts to affect our progress, we ask the TA for guidance, and the instructor is the final step. Concerns about how members work together stay inside the group until we reach the TA or instructor step. We focus on the issue and not on blaming people, and we check back one week after any agreement to see whether it worked. When a concern is about project expectations or the work on site, we may also consult Dr. Dudley. A conflict with a CPC or partner is raised with the group first and then handled by Hadar politely and in writing, or by Joseph if the conflict involves Hadar. Any agreed change in responsibilities is shared with everyone affected.

I. Signatures

By signing, each member confirms that they have read this contract, agree to it, and will follow it for the whole project.

Mehrshid Abdoli Signature [sign here] Date [date]

Hadar Al-Ali Signature [sign here] Date [date]

Joseph DeGiorgi Signature [sign here] Date [date]

Diya Jain Signature [sign here] Date [date]

Marchelino Toma Signature [sign here] Date [date]

7. Risks and Mitigations

We found nine risks that could realistically set this project back. For each one we explain what could go wrong, how likely it is, and who is watching for it, and then we list what we will do about it.

1. Weather or frozen ground on planting day

Heavy rain, snow, or soaked clay could force us to postpone seeding, and a downpour right after sowing could wash seed off the plot. We rate this risk as medium likelihood and high impact, and Mehrshid owns it.

Mitigation. We plant on Saturday, November 21 so that we still have a backup date, Saturday, November 28, and we tell volunteers about both dates from the start. From November 16 we check the forecast every day and make a written decision on Wednesday, November 18. We postpone if more than about 10 mm of rain falls in the two days before, if the clay sticks to boots, if winds above 30 km/h are forecast, or if snow covers the plot. Light frost alone does not stop us, because seed sown in the dormant season is meant to sit on cold soil. We cover the seed with a thin layer of straw straight after sowing, and we keep the layer thin on purpose, because light straw helped some prairie seedlings emerge while heavy straw held others back (Mollard et al., 2014). If November 28 also fails, we give the weighed and labelled seed bags and written instructions to Nature @ McMaster so they can sow when conditions allow, and we explain this in the final report. If we plant on November 28, the baseline record moves to Thursday, December 3 and the report draft keeps its date.

2. Delays or changes in the site, plot preparation, or partner communication

The site could change after Dr. Dudley talks with grounds staff, Nature @ McMaster and Facility Services may not have the plot ready in time, or with our CPC, Nature @ McMaster, and a grounds team involved we could miss an instruction or receive two that conflict. We rate this risk as medium likelihood and high impact, and Hadar owns it.

Mitigation. We draw the layout so it can shift to another open area without redesigning the experiment. We confirm the site and plot size in writing at the site visit by October 23, and we ask for plot readiness in writing on November 9 and again on November 16. Hadar sends all email to Dr. Dudley and Nature @ McMaster with every member copied, any member who speaks with a partner or supplier reports back in Teams the same day, meeting notes go to Dr. Dudley within 24 hours, and we ask for a written yes on any change to the site, species, budget, or dates. If the plot is not ready by November 18, we move to the November 28 backup date, which Dr. Dudley will already have approved in week 8.

3. Seed cost, availability, delivery, or quality

Native seed can sell out by late fall, it may cost more than the project budget can cover, delivery could be slow, and some wildflower mixes sold as native still contain plants that are not. We rate this risk as medium likelihood and medium impact, and Marchelino owns it.

Mitigation. At the October 14 meeting we ask Dr. Dudley for a budget figure and for who places and pays for the order, so that no member pays out of pocket and the cost score in the framework uses real limits. Seed availability and cost are part of the evaluation framework, so we avoid species we cannot buy. We ask at least two suppliers for quotes, stock, and delivery dates by October 30, and we read every species list on a mix label and reject mixes that include plants that are not native to Ontario. We order by Wednesday, November 4 and ask for delivery or local

pickup by November 13. If the seed has not arrived by Monday, November 16, Marchelino proposes the closest substitute from the ranked backup list in D2, and once the group and Dr. Dudley agree in writing, it is bought from a local supplier, or we move to the November 28 date. If money runs short, we shrink the subplots before dropping a mix, so all three mixes are still tested.

4. Helicopter activity and safety near the helipad

Rotor wash from landing helicopters could blow seed and straw off the plot, loose straw could become debris near the landing area, and work could be stopped during a landing. Volunteers working on site may also need approval from McMaster. We rate this risk as medium likelihood and high impact, and Mehrshid owns it.

Mitigation. At the site visit we ask Facility Services and Dr. Dudley how far the plot must be from the landing area, whether loose straw is allowed there, and who we contact on planting day. If loose straw is not allowed, we ask Dr. Dudley about a biodegradable erosion blanket pinned over the seed instead. By October 30 we also ask whether volunteers need a waiver or any McMaster approval. On planting day one member watches for helicopters, and everyone moves off the plot until each landing is over. Work runs from 10 a.m. to 1 p.m. so that we finish well before dark, and the planting plan lists warm clothing, gloves, and closed footwear for everyone.

5. Clay soil and site conditions limit establishment

Clay that stays wet, compacts, or crusts could stop seedlings from emerging in spring, and some species on our list may not cope with these conditions. We rate this risk as medium likelihood and medium impact, and Marchelino owns it.

Mitigation. Clay tolerance carries one of the highest weights in the evaluation framework, so the shortlist favours species that can handle it. Mehrshid's site walk on October 19 records drainage, standing water, and bare or compacted patches, and we ask Nature @ McMaster at the site visit how the soil will be loosened during preparation. We do not plan to water over winter, since dormant seed relies on snowmelt and spring rain, but the monitoring protocol tells the spring team when and how to water if May is dry.

6. Foot traffic and winter damage to the plot

People may cut across the plot, and mowers, snowplows, snow piles, or road salt could disturb it over winter, which would damage seedlings and mix up the results. We rate this risk as medium likelihood and medium impact, and Diya owns it.

Mitigation. We place the plot away from the paths people already use, which we check at the site walk. With approval from Dr. Dudley and Facility Services, we mark the plot with stakes and rope and a small sign that explains the project. We mark every subplot corner with metal stakes set flush with the ground and record GPS points, so the subplots can be found even if the surface markers are removed. We also ask Facility Services in writing to keep mowers, snow piles, and salt off the plot until spring.

7. Weeds and invasive species outcompete the seedlings

Weeds already in the soil, or seed carried in on straw, could sprout in spring and crowd out the native seedlings, which would also make the weed cover counts unreliable. We rate this risk as high likelihood and medium impact, and Joseph owns it.

Mitigation. We ask Dr. Dudley for clean straw that is free of weed and grain seed, and we only buy seed with a full species list. Joseph researches the main weed threats for these mixes as part of the literature review, and the monitoring protocol includes weed cover and a simple spring weeding schedule that we ask Nature @ McMaster to follow. The turf control subplots also show how much weed pressure the site has without any change in planting.

8. Results cannot be seen before December 5

Seed sown in late November will not germinate until spring, so the term ends before we can see whether any mix worked. We rate this risk as certain and medium impact, and Diya owns it.

Mitigation. We define success for this term as a replicated plot, a documented baseline, and a ready monitoring protocol. We record plot maps, dated photos, GPS points, and the sowing record so spring results can be compared with the start. We ask Dr. Dudley at the November 11 meeting who will monitor in 2027, and we hand the package to that person or to Dr. Dudley by December 4. All files go into a folder owned by Dr. Dudley so they stay available after the term.

9. Workload clashes, low turnout, or a member unavailable

November is busy in our other courses, a member could be sick on planting weekend, and fewer volunteers than expected could show up. We rate this risk as high likelihood and medium impact, and Hadar owns it.

Mitigation. Each member posts their November test and assignment dates in the timeline tracker by Friday, October 16. At each weekly meeting we look two weeks ahead, and when a member has two or more other deadlines in one week, we move one of their tasks to another member and record the swap in the contribution log. Each lead writes an outline of their report section by November 13, so the writing in weeks 10 and 11 starts from a plan. For planting day we invite three to five volunteers, which is plenty for twelve small subplots in three hours, and we send reminders one week, two days, and one day ahead. Only group members broadcast seed, while volunteers rake, carry straw, and take photos. The planting plan lists every task with a time and a person, so the work can go ahead even if one member is missing.

8. Project Plan Submission to CPCs

The completed plan was saved as a PDF and emailed to our CPC, Dr. Susan Dudley, with all five group members copied and the plan attached as a PDF.

The screenshot below shows the recipients, the copied members, the date and time sent, and the attachment.

[Insert a screenshot of the sent email here. It must show Dr. Dudley as the recipient, all five members copied, the date and time sent, and the plan attached as a PDF.]

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